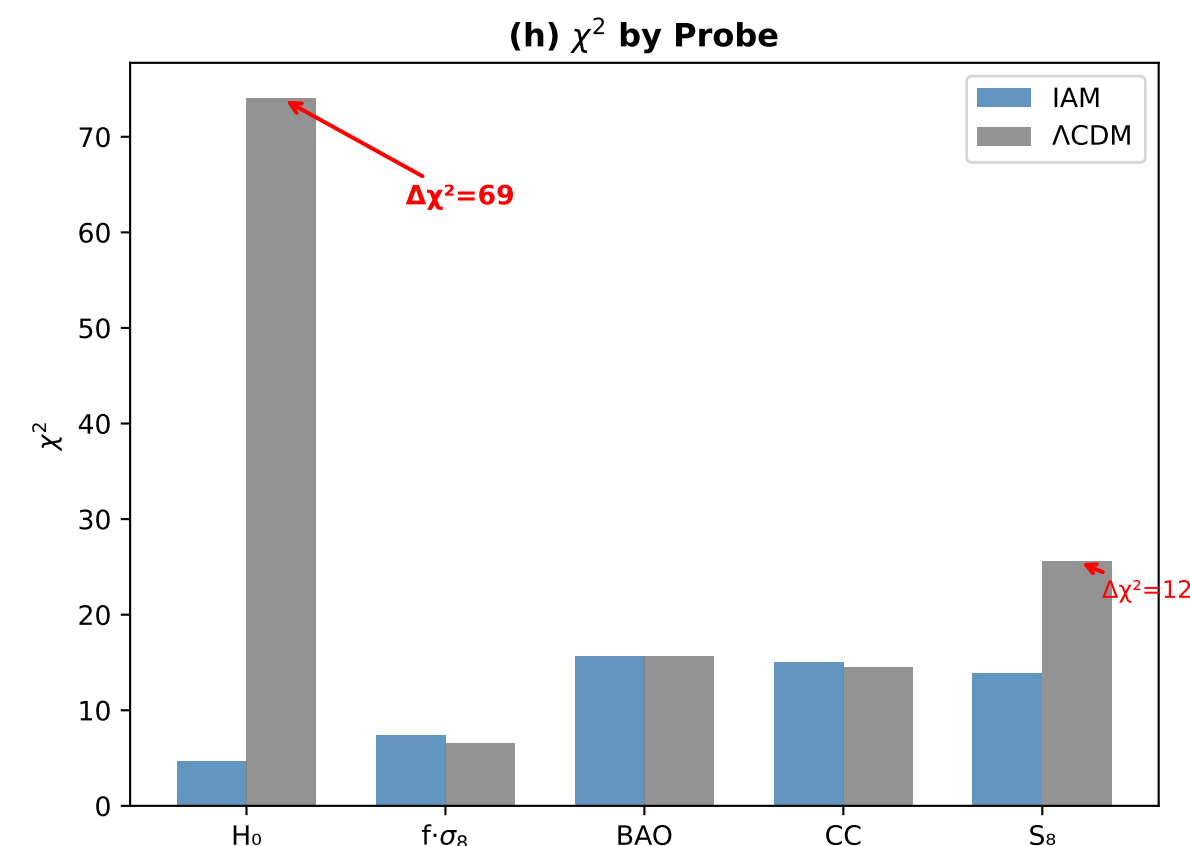
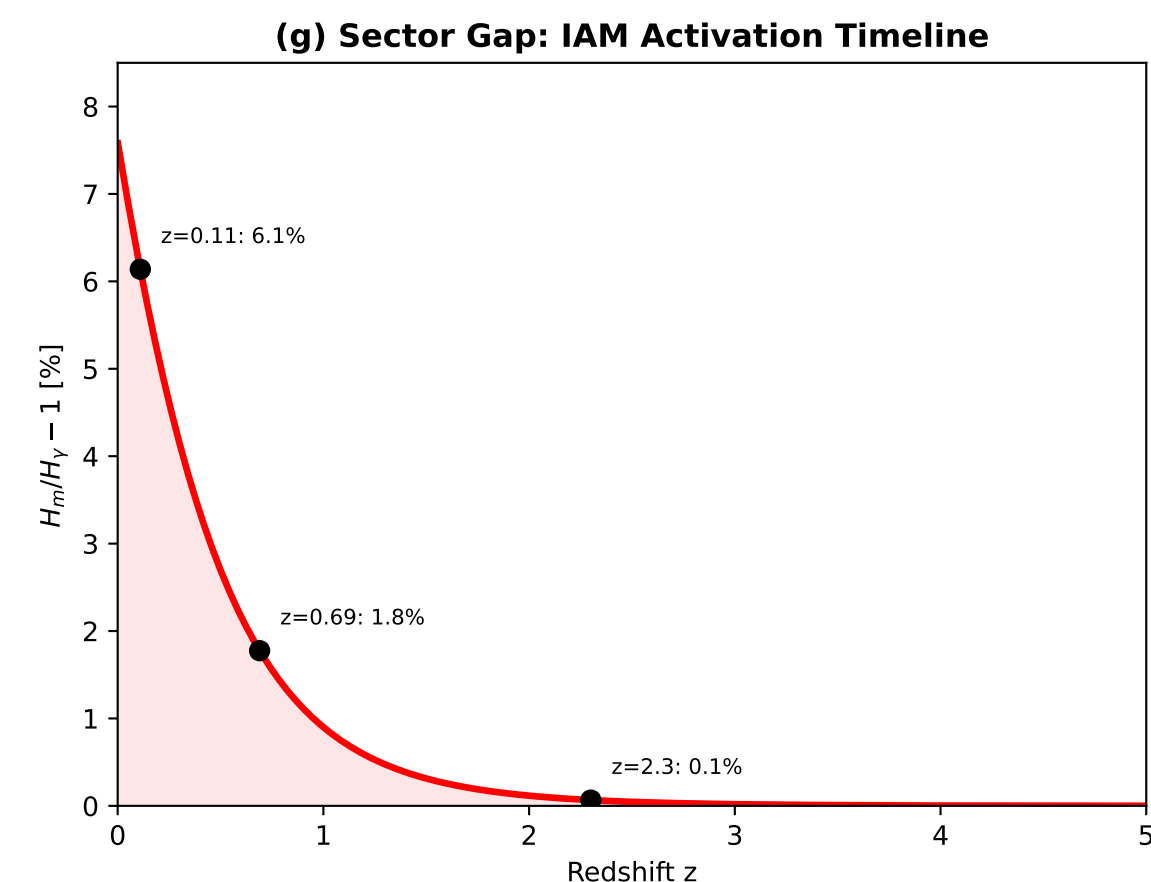
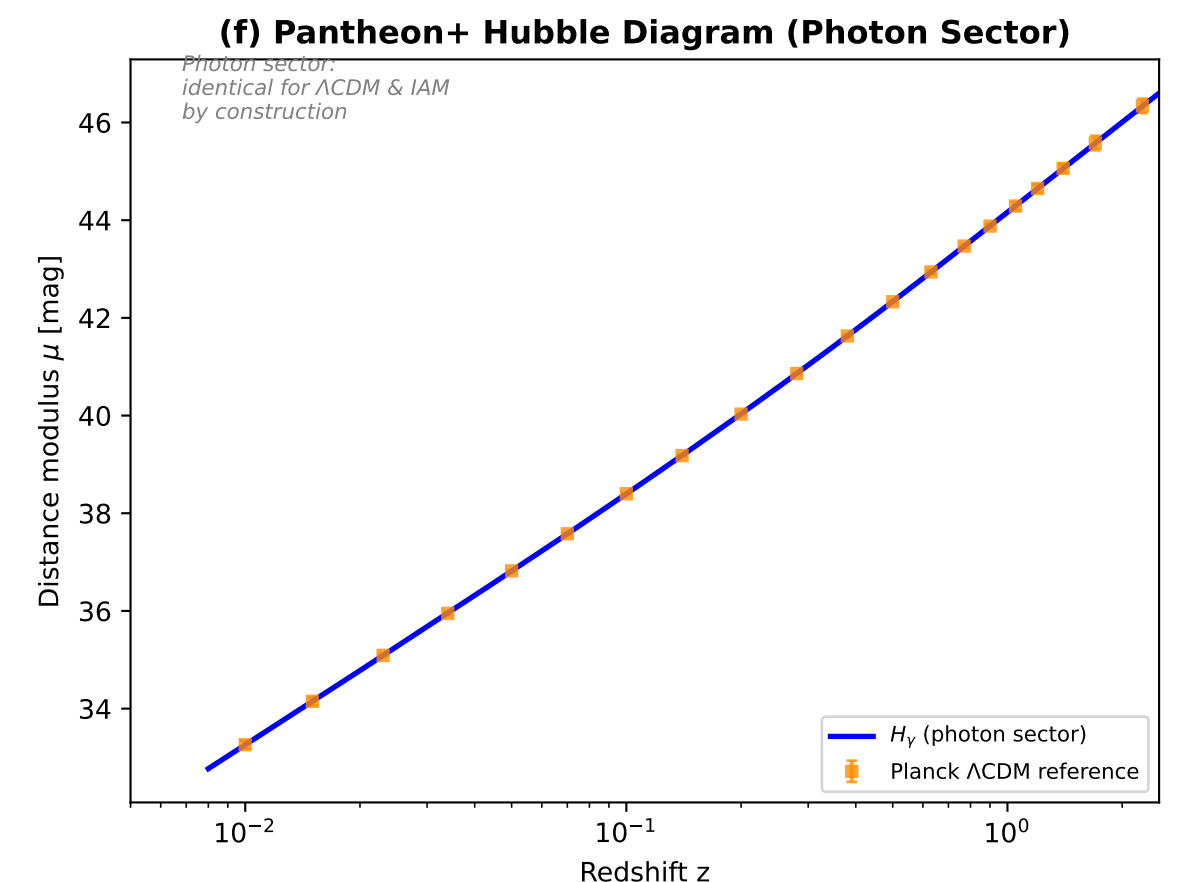
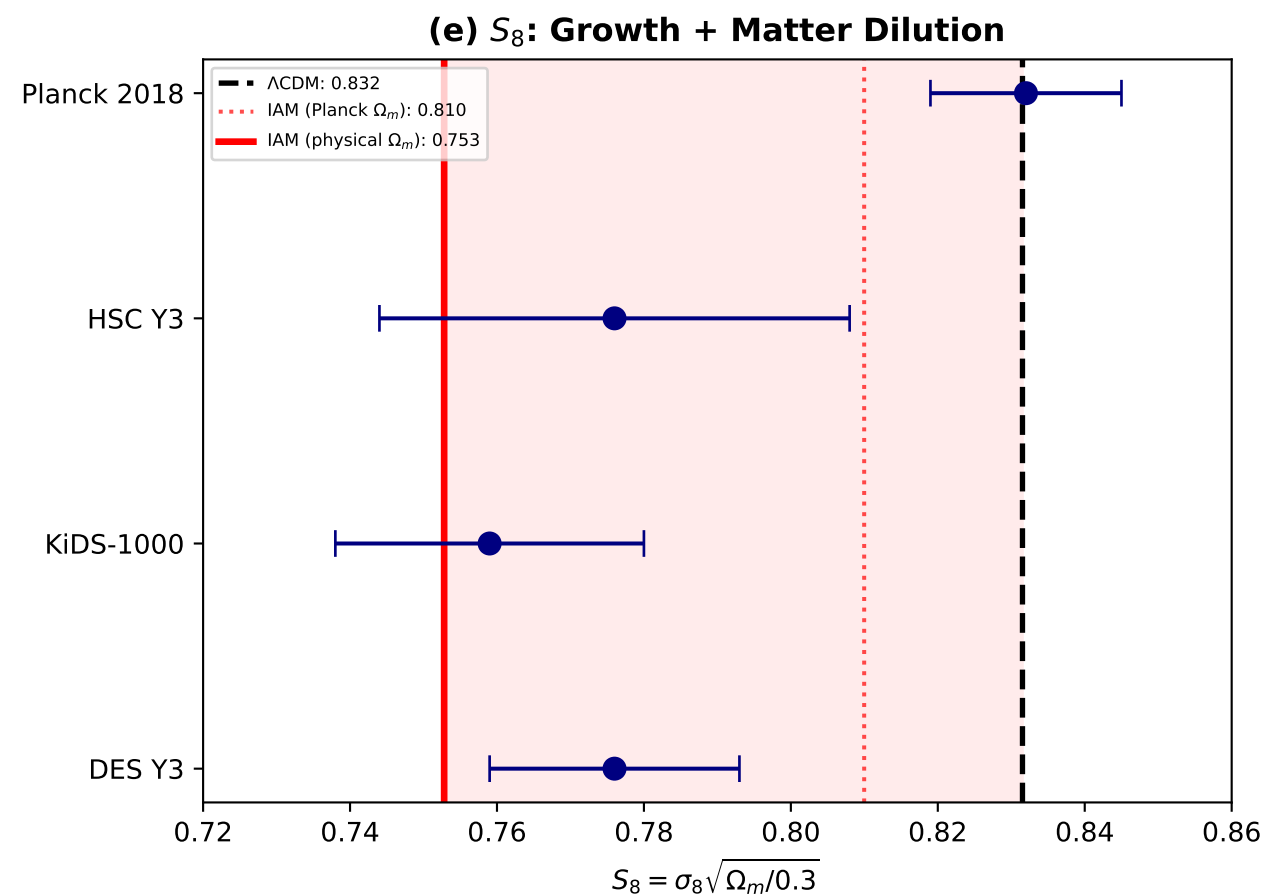
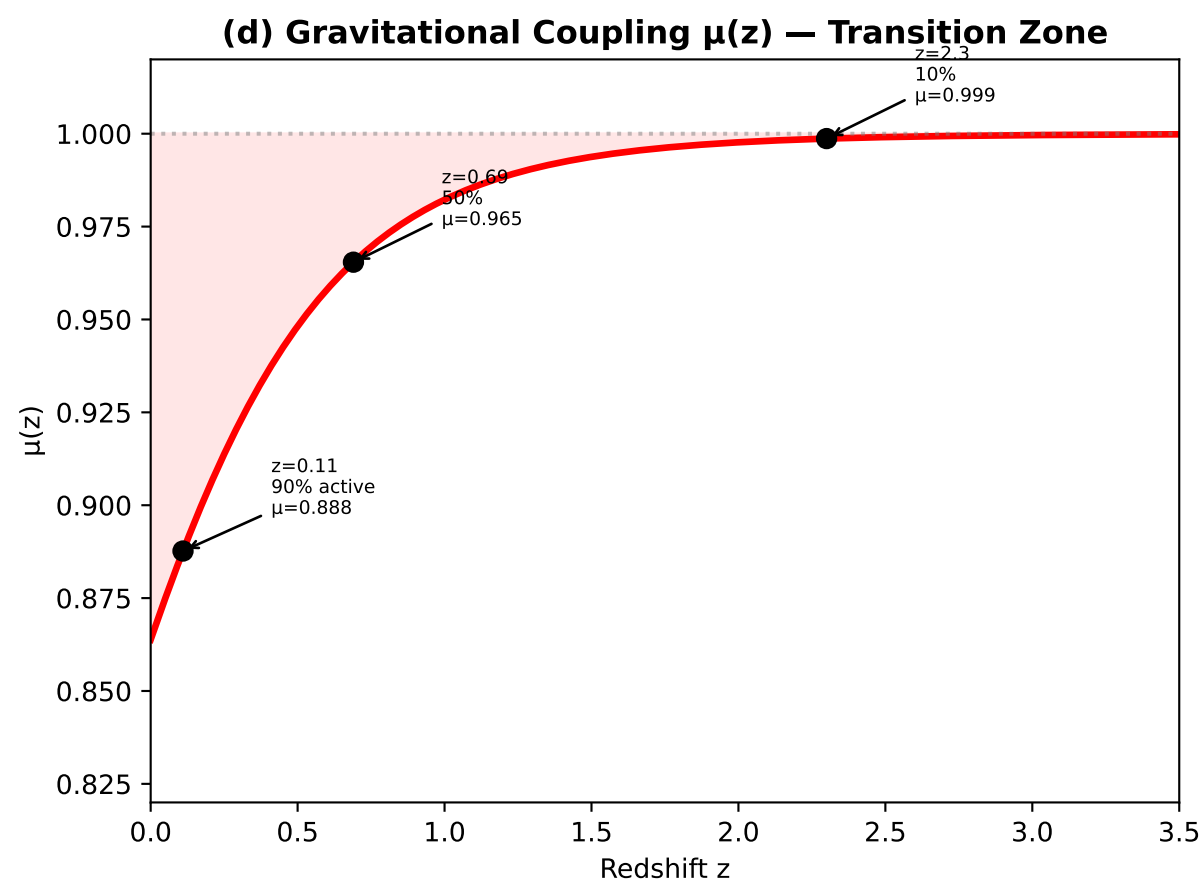
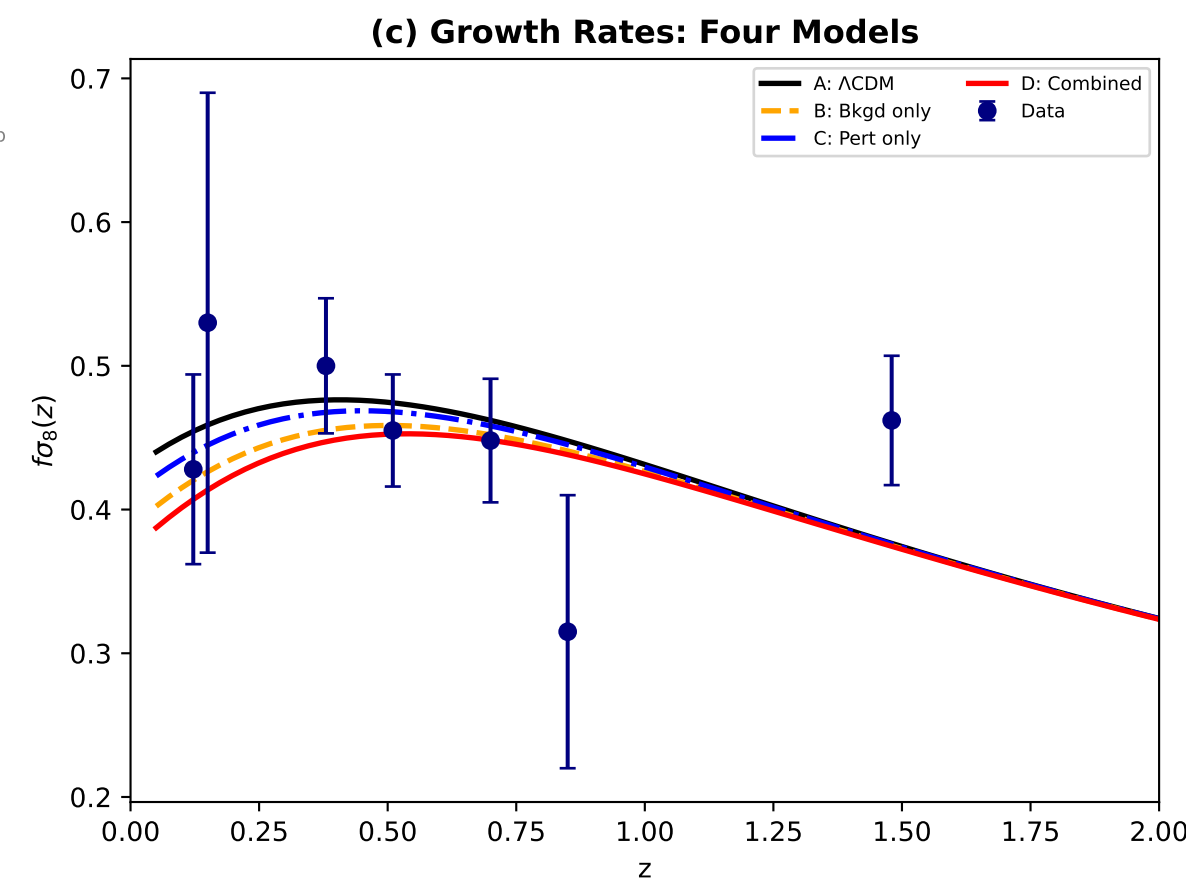
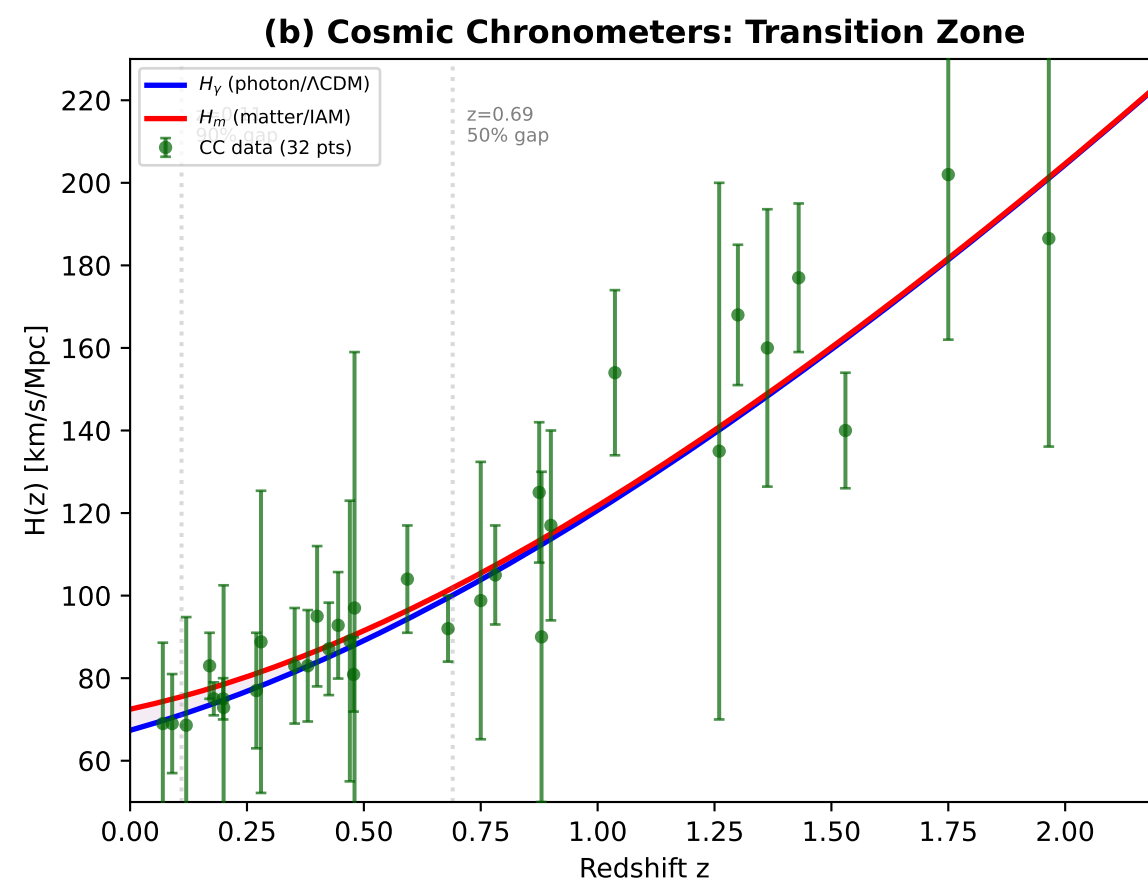
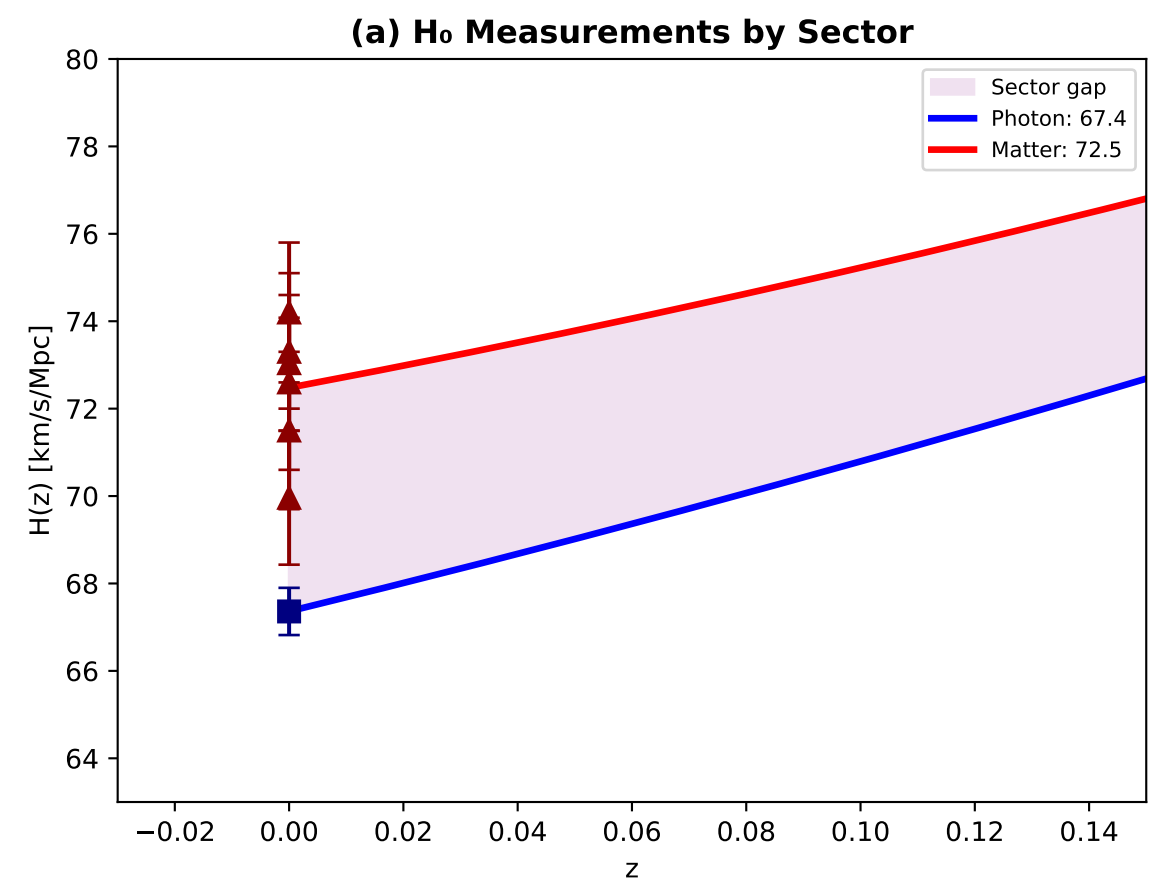


# IAM Dual-Sector Validation: 65 Data Points Across 6 Probes (+ Pantheon+ Consistency Check)

## Combined $\Delta\chi^2 = 79.8$ (equivalent to $8.9\sigma$ ), zero additional parameters



**IAM Dual-Sector Validation Summary**

beta =  $0m/2 = 0.15765$  (virial theorem)  
 $H_0(\text{photon}) = 67.36$  km/s/Mpc  
 $H_0(\text{matter}) = 72.48$  km/s/Mpc  
 $\mu(z=0) = 0.8638$   
 $\sigma_8(\text{IAM}) = 0.7901$  (2.6% below LCDM)  
 $S_8(\text{IAM, Planck } 0m) = 0.810$   
 $S_8(\text{IAM, physical } 0m=0.272) = 0.753$

chi2 comparison: 65 points, 6 probes  
 $H_0$ : 7 RSD: 7 BAO: 12  
CC: 32  $S_8$ : 4 CMB: 3  
Plus 20-bin Pantheon+ (photon sector, identical by construction)

dchi2(LCDM - IAM) = 79.8  
Equivalent significance: 8.9 sigma  
Additional free parameters: 0

Transition zone (E(a) activation):  
90% at  $z = 0.11$   
50% at  $z = 0.69$   
10% at  $z = 2.3$   
GR recovered for  $z > 5$